

Great! Let's go through a **refresher on Differential Calculus** step by step. 🚀

1. What is Calculus?

Calculus is the branch of mathematics that studies **change**. It has two major parts:

1. **Differential Calculus** – Deals with the rate of change (slopes, derivatives).
2. **Integral Calculus** – Deals with accumulation (areas, summing small parts).

Since we are focusing on **Differential Calculus**, let's dive into it!

2. Understanding Slope & Rate of Change

Before calculus, we use the concept of **slope** in algebra to measure how steep a line is.

For a straight line:

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

But what if we have a **curve** (like $y = x^2 + 2$)? The slope is **not constant**! Instead, it **changes** at every point.

3. Instantaneous Rate of Change (Derivative)

To find the slope of a curve at a specific point, we use **limits**.

1. Pick two points on the curve:

- One at x , which has value $f(x)$.
- Another close by at $x + h$, which has value $f(x + h)$.

2. Compute the average rate of change (slope between two points):

$$\frac{f(x+h) - f(x)}{h}$$

3. Take the limit as $h \rightarrow 0$:

- This brings the second point infinitely close to the first, giving the **instantaneous slope**.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This is called the **derivative** of $f(x)$, denoted as $f'(x)$ or $\frac{dy}{dx}$.

4. Basic Rules of Differentiation

Once we understand the definition, we can use **rules** to differentiate functions quickly:

1. Power Rule:

$$\frac{d}{dx} x^n = nx^{n-1}$$

Example: $\frac{d}{dx} x^3 = 3x^2$

2. Constant Rule:

$$\frac{d}{dx} (C) = 0$$

Example: $\frac{d}{dx} (5) = 0$

3. Sum/Difference Rule:

$$\frac{d}{dx} (f(x) + g(x)) = f'(x) + g'(x)$$

Example: $\frac{d}{dx}(x^3 + 2x) = 3x^2 + 2$

4. **Product Rule:**

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

5. **Quotient Rule:**

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

6. **Chain Rule:**

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Used for composite functions.

5. **Example: Finding the Derivative of $y = x^2 + 2$**

Using the **power rule**:

$$\frac{d}{dx}(x^2 + 2) = 2x + 0 = 2x$$

This means the **instantaneous slope** of $y = x^2 + 2$ at any point x is given by $2x$.

To find the slope at $x = 5$:

$$f'(5) = 2(5) = 10$$

So, the slope at $x = 5$ is **10**.

6. Why is This Useful?

1. **Physics** – Derivatives measure velocity, acceleration, and other rates of change.
2. **Economics** – Used in cost functions, profit maximization, etc.
3. **Machine Learning & AI** – Optimization algorithms rely on derivatives.
4. **Engineering** – Used in control systems, signal processing, and mechanics.

7. Next Steps

- If you want more practice, try differentiating:

i. $f(x) = x^3 + 4x$

ii. $g(x) = \frac{1}{x}$

iii. $h(x) = \sin x$

Would you like a deeper dive into any specific topic? 

1. To find the slope of the function

To find the slope of the function $y = x^2 + 2$ at $x = 5$, follow these steps:

1. **Find the derivative of y :**

The derivative of $y = x^2 + 2$ gives the slope of the function at any point x .

$$\frac{dy}{dx} = \frac{d}{dx}(x^2 + 2)$$

Using the power rule $\frac{d}{dx}x^n = nx^{n-1}$, we get:

$$\frac{dy}{dx} = 2x$$

2. **Evaluate the derivative at $x = 5$:**

$$\left. \frac{dy}{dx} \right|_{x=5} = 2(5) = 10$$

So, the slope of the function at $x = 5$ is **10**.